Department of Information Technology

Academic Year 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3391 & Object-Oriented Programming

Name of the Faculty member (s): Dr. K. Palraj, ASP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / User Defined Exceptions
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO11
- **Activity Chosen: Mind map**
- **Justification:**
    - A mind map is an excellent way to represent the concept of user-defined exceptions because it provides a clear and organized visual structure. It allows learners to see how concepts like custom class creation, exception hierarchy, throw and catch blocks, and constructors are all interconnected.
    - Mind maps help in simplifying complex topics, making it easier to understand the flow of creating and handling custom exceptions. They improve memory retention as students can recall the structure visually.
    - This format highlights relationships between key components, encourages active participation by letting learners add examples and notes, and serves as a quick revision tool before exams
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
    - The mind map was created to visually organize the topic in a structured and easy-to-understand format. The central theme was placed at the centre of the diagram, and connected branches were drawn to represent the major subtopics.
    - Each branch was further divided into smaller nodes to include supporting ideas, key points, and examples. This structure helps present the information in a logical flow and makes it easy to understand the relationships between different concepts.

- **Images / Screenshot of the practice:**

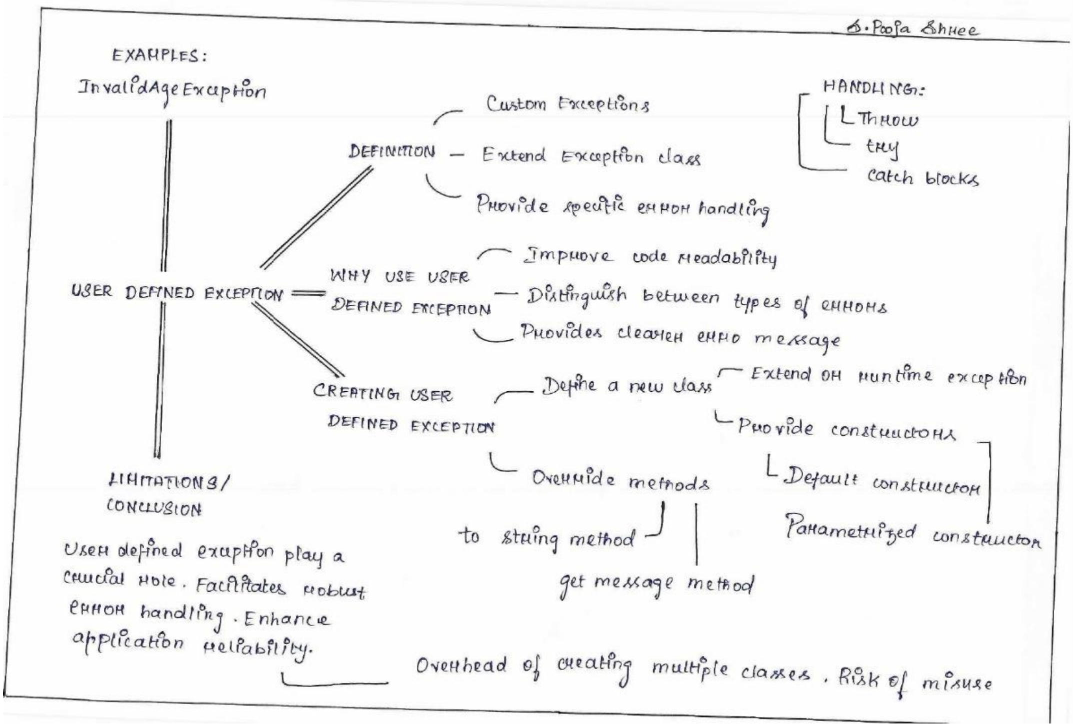

Figure 1: Mindmap User Defined Exception of S. Pooja Shree

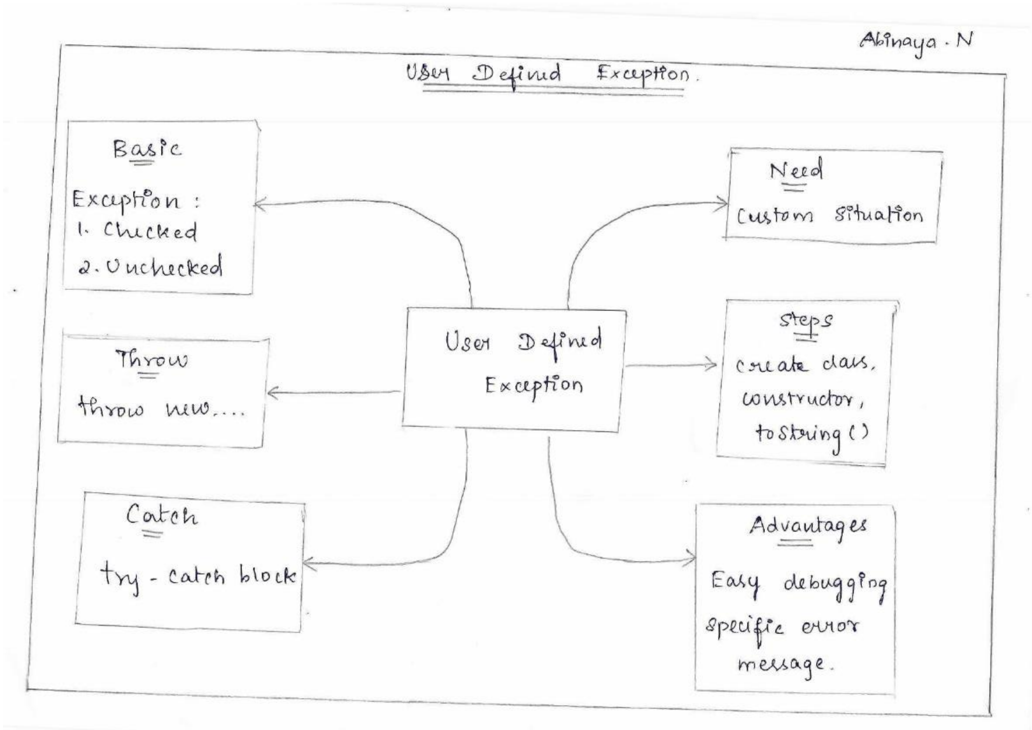

Figure 2: ***Mindmap User Defined Exception of N. Abinaya***

- ***CO – PO / PSO mapping:***

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	3	2	3	3	3	1	1	3	2	2

(1 – Low 2 – Moderate 3 – High)

- ***PO / PSO mapped:***

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	3	3	3	2	3	3	3	1	1	3	2	2
Justification for correlation	Students will be able to apply core Java concepts to develop robust multithreaded programs using exception handling	Students will be able to analyze real-world situations where concurrency or error handling is needed, and apply suitable techniques	Students will be able to write programs using threads and exception handling require thoughtful structuring to ensure reliability and responsiveness	Students will be able to test and analyze multithreaded programs to ensure they function correctly under different conditions	Students use IDEs, debuggers, and compilers (e.g., Eclipse, IntelliJ) to write and test object-oriented programs	Students will be able to involve designing and implementing concurrent systems collaboratively in the group tasks	Students will be able to explain how multithreading and exception handling improve application performance and stability	Students will be able to plan and manage multithreaded application development phases in assignments or projects.	Students could be able to learning advanced Java features like threads and exceptions encourages further exploration in concurrent and secure programming	Students will be able to handles errors and manages parallel tasks in data-intensive or simulation-based applications	Multithreading and exception handling are key in real-time systems, embedded applications, IoT, and server-side development	Students will be able to development of fault-tolerant and high-performing software components, following software engineering practices

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Students reported that creating and handling user-defined exceptions helped them understand exception handling.
    - Some students suggested more examples of real-life scenarios where custom exceptions are applied.
    - The exercises improved their coding confidence and ability to implement custom error handling.
    - Observed that students gained better clarity on how user-defined exceptions integrate with Java's exception hierarchy

  - ❖ ***Benefit of the practice:***
    - Students learn how to create custom exceptions and manage errors beyond predefined Java exceptions.
    - Practicing user-defined exceptions strengthens programming logic, syntax knowledge, and debugging skills.
    - Engaging exercises, examples, and scenarios keep learners motivated and encourage critical thinking

  - ❖ ***Challenges faced in implementation:***
    - Some students found it difficult to grasp the difference between built-in exceptions and user-defined exceptions initially.
    - Creating a custom exception class and properly extending Exception or RuntimeException caused confusion for beginners.
    - Students sometimes struggled to identify real-world scenarios where custom exceptions would be useful.

References:

❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/4.Mindmap%20Unit%204.pdf

❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/CS3391_UNIT%202_%20MM_SS.pdf

❖ https://equiip.eu/activity/activity-5-mind-mapping/

Signature of Faculty Member **HOD**